Decision Tree

Unit - 3

Decision tree

- Decision tree induction is a supervised learning method used for classification and prediction.
- It recursively splits data based on attribute values to form a tree structure.

- **How are decision trees used for classification?**
- Classification is done by **traversing** the tree from the root to a leaf:
  - At each internal node, test the attribute associated with the node.
  - Follow the branch that matches the attribute value in the input tuple.
  - Repeat until a leaf is reached → the class label at that leaf is the prediction.

Why are decision tree classifiers so popular?

- **Easy to understand and interpret** (visualization as flowcharts).
- **No domain knowledge** required in advance.
- Can handle both **numerical and categorical data**.
- Perform well on **large datasets**.
- Inexpensive to build and fast to classify.
- Support **white-box models** (transparent logic, not a black box).

- A **splitting attribute** is the attribute (feature) chosen at a particular node in a decision tree to divide (split) the dataset into subsets.

- To partition the data such that the resulting subsets are more "pure" (i.e., contain mostly tuples of a single class).

- The goal is to make the decision tree learn patterns that help in classifying new data accurately.

- **How is it chosen?**
  - Using **attribute selection measures** such as:
    - **Information Gain**
    - **Gain Ratio**
    - **Gini Index**

- **Splitting Attributes: Discrete-Valued vs Continuous-Valued**

- **Discrete-valued** attributes:
  - Have a finite number of values (e.g., Color = {Red, Green, Blue}).
  - Splitting is done by creating branches for each distinct value.
  - May allow **multiway splits** (more than binary branches).

- **Continuous-valued** attributes:
  - Real-number attributes (e.g., Age, Income).
  - A **split-point** (e.g., Age ≤ 30) is selected based on information gain.
  - Partitioned into two intervals (binary split usually): ≤ split-point and > split-point.

- **Concept of Majority Voting**

- Used when:
- All attributes have been used, but not all tuples belong to the same class.

- Some data partitions are empty.

- **Majority voting** assigns the class that appears most frequently in the current subset.
- **Example**: In a node with class labels: [Yes, Yes, No, Yes] → majority = **Yes** → assign Yes.

Algorithm: Generate_decision_tree. Generate a decision tree from the training tuples of data partition, *D*.

Input:

- Data partition, *D*, which is a set of training tuples and their associated class labels;
- *attribute_list*, the set of candidate attributes;
- *Attribute_selection_method*, a procedure to determine the splitting criterion that "best" partitions the data tuples into individual classes. This criterion consists of a *splitting_attribute* and, possibly, either a *split-point* or *splitting subset*.

Output: A decision tree.

Method:

(1) create a node *N*;
(2) **if** tuples in *D* are all of the same class, *C*, **then**
(3)   return *N* as a leaf node labeled with the class *C*;
(4) **if** *attribute_list* is empty **then**
(5)   return *N* as a leaf node labeled with the majority class in *D*; // majority voting
(6) apply **Attribute_selection_method**(*D*, *attribute_list*) to **find** the "best" *splitting_criterion*;
(7) label node *N* with *splitting_criterion*;
(8) **if** *splitting_attribute* is discrete-valued **and**
    multiway splits allowed **then** // not restricted to binary trees
(9) $attribute\_list \leftarrow attribute\_list - splitting\_attribute$; // remove *splitting_attribute*
(10) **for each** outcome *j* of *splitting_criterion*
// partition the tuples and grow subtrees for each partition
(11)   let D_j be the set of data tuples in *D* satisfying outcome *j*; // a partition
(12)   **if** D_j is empty **then**
(13)   attach a leaf labeled with the majority class in *D* to node *N*;
(14)   **else** attach the node returned by **Generate_decision_tree**(D_j, *attribute_list*) to node *N*;
endfor
(15) return *N*;

Basic Decision Tree Algorithm

Day	Outlook	Temp	Humidity	Wind	Play Tennis
D1	Sunny	Hot	High	Weak	No
D2	Sunny	Hot	High	Strong	No
D3	Overcast	Hot	High	Weak	Yes
D4	Rain	Mild	High	Weak	Yes
D5	Rain	Cool	Normal	Weak	Yes
D6	Rain	Cool	Normal	Strong	No
D7	Overcast	Cool	Normal	Strong	Yes
D8	Sunny	Mild	High	Weak	No
D9	Sunny	Cool	Normal	Weak	Yes
D10	Rain	Mild	Normal	Weak	Yes
D11	Sunny	Mild	Normal	Strong	Yes
D12	Overcast	Mild	High	Strong	Yes
D13	Overcast	Hot	Normal	Weak	Yes
D14	Rain	Mild	High	Strong	No

$$Info(D) = -\sum_{i=1}^{m} p_i \log_2(p_i),$$

$$Info_A(D) = \sum_{j=1}^{v} \frac{|D_j|}{|D|} \times Info(D_j).$$

$$Gain(A) = Info(D) - Info_A(D).$$

Information Gain

Day	Outlook	Temp	Humidity	Wind	Play Tennis
D1	Sunny	Hot	High	Weak	No
D2	Sunny	Hot	High	Strong	No
D3	Overcast	Hot	High	Weak	Yes
D4	Rain	Mild	High	Weak	Yes
D5	Rain	Cool	Normal	Weak	Yes
D6	Rain	Cool	Normal	Strong	No
D7	Overcast	Cool	Normal	Strong	Yes
D8	Sunny	Mild	High	Weak	No
D9	Sunny	Cool	Normal	Weak	Yes
D10	Rain	Mild	Normal	Weak	Yes
D11	Sunny	Mild	Normal	Strong	Yes
D12	Overcast	Mild	High	Strong	Yes
D13	Overcast	Hot	Normal	Weak	Yes
D14	Rain	Mild	High	Strong	No

- **Values(OUTLOOK) = {Sunny, Overcast, Rain}**
- $S = \{ 9 , 5 \}$
- $S_{SUNNY} = \{ 2 , 3 \}$
- $S_{OVERCAST} = \{ 4 , 0 \}$
- $S_{RAIN} = \{ 3 , 2 \}$

Information Gain

Day	Outlook	Temp	Humidity	Wind	Play Tennis
D1	Sunny	Hot	High	Weak	No
D2	Sunny	Hot	High	Strong	No
D3	Overcast	Hot	High	Weak	Yes
D4	Rain	Mild	High	Weak	Yes
D5	Rain	Cool	Normal	Weak	Yes
D6	Rain	Cool	Normal	Strong	No
D7	Overcast	Cool	Normal	Strong	Yes
D8	Sunny	Mild	High	Weak	No
D9	Sunny	Cool	Normal	Weak	Yes
D10	Rain	Mild	Normal	Weak	Yes
D11	Sunny	Mild	Normal	Strong	Yes
D12	Overcast	Mild	High	Strong	Yes
D13	Overcast	Hot	Normal	Weak	Yes
D14	Rain	Mild	High	Strong	No

- **Values(OUTLOOK) = {Sunny, Overcast, Rain}**

- **S = { 9 , 5 }**
- **Info(S) = [- (9/14) $\log_2$(9/14)] + [- (5/14) $\log_2$(5/14)]**
- **Info(S) = 0.94**

- **S_{SUNNY} = { 2 , 3 }**
- **Info(S_{SUNNY}) = [- (2/5) $\log_2$(2/5)] + [- (3/5) $\log_2$(3/5)]**
- **Info(S_{SUNNY}) = 0.971**

- **$S_{OVERCAST}$ = { 4 , 0 }**
- **Info($S_{OVERCAST}$) = [- (4/4) $\log_2$(4/4)] + [- (0/4) $\log_2$(0/4)]**
- **Info($S_{OVERCAST}$) = 0**

- **S_{RAIN} = { 3 , 2 }**
- **Info(S_{RAIN}) = [- (3/5) $\log_2$(3/5)] + [- (2/5) $\log_2$(2/5)]**
- **Info(S_{RAIN}) = 0.971**

- **Values(OUTLOOK) = {Sunny, Overcast, Rain}**

$$Gain(S, Outlook) = Entropy(S) - \sum_{v \in \{Sunny, Overcast, Rain\}} \frac{|S_v|}{|S|} Entropy(S_v)$$

- **S = { 9 , 5 }**
- **Info(S) = 0.94**

- **S_{SUNNY} = { 2 , 3 }**
- **Info(S_{SUNNY}) = 0.971**

- **$S_{OVERCAST}$ = { 4 , 0 }**
- **Info($S_{OVERCAST}$) = 0**

- **S_{RAIN} = { 3 , 2 }**
- **Info(S_{RAIN}) = 0.971**

- Gain (S, Outlook) =

0.94 - { [(5/14) * Info(S_{SUNNY})]

+ [(4/14) * Info($S_{OVERCAST}$)]

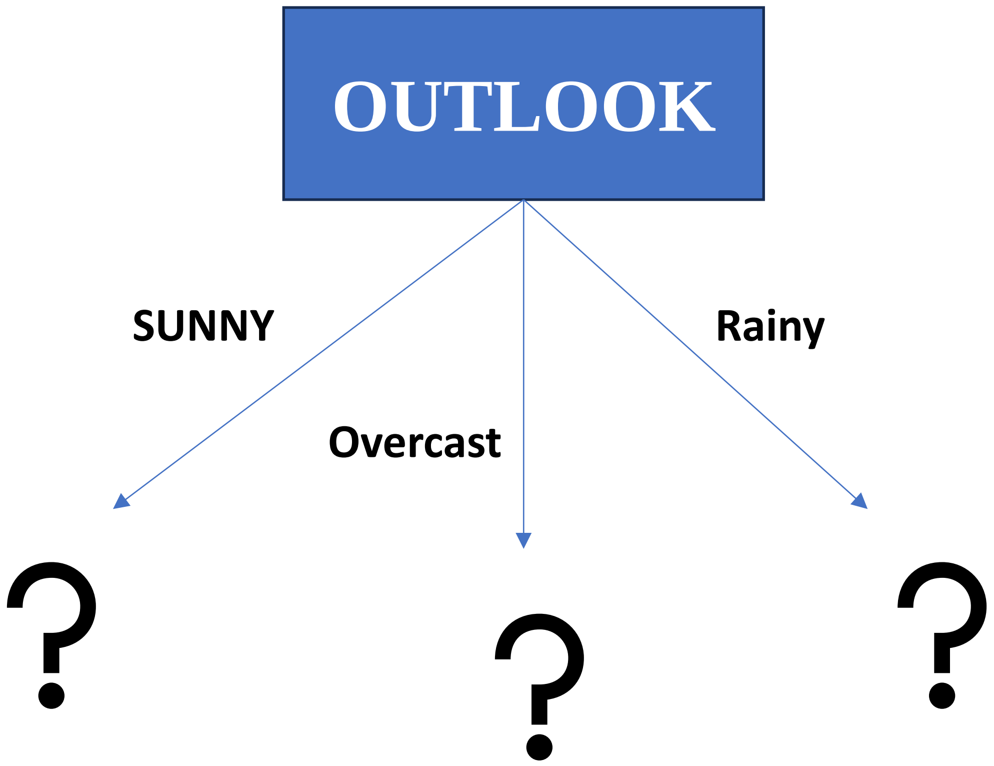
+ [(5/14) * Info(S_{RAIN})]

}

- Gain (S, Outlook) = 0.2464

Similarly compute Gain for other attributes

- Gain (S, Outlook) = 0.2464
- Gain (S, Temp) = 0.0289
- Gain (S, Humidity) = 0.1516
- Gain (S, Wind) = 0.0478



Day	Outlook	Temp	Humidity	Wind	Play Tennis
D1	Sunny	Hot	High	Weak	No
D2	Sunny	Hot	High	Strong	No
D3	Overcast	Hot	High	Weak	Yes
D4	Rain	Mild	High	Weak	Yes
D5	Rain	Cool	Normal	Weak	Yes
D6	Rain	Cool	Normal	Strong	No
D7	Overcast	Cool	Normal	Strong	Yes
D8	Sunny	Mild	High	Weak	No
D9	Sunny	Cool	Normal	Weak	Yes
D10	Rain	Mild	Normal	Weak	Yes
D11	Sunny	Mild	Normal	Strong	Yes
D12	Overcast	Mild	High	Strong	Yes
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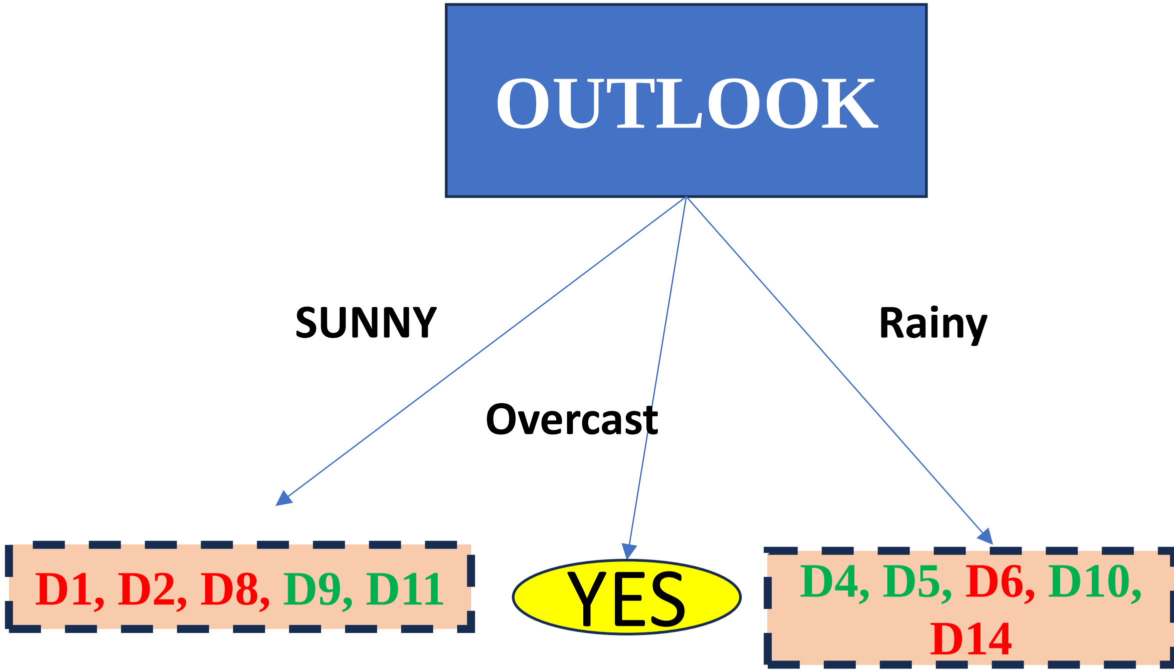


Day	Temp	Humidity	Wind	Play Tennis
D1	Hot	High	Weak	No
D2	Hot	High	Strong	No
D8	Mild	High	Weak	No
D9	Cool	Normal	Weak	Yes
D11	Mild	Normal	Strong	Yes

These rows contain data when Outlook is SUNNY. Find the Gain of Temp, Humidity, Wind wrt Sunny.

- Gain ($S_{Outlook}$, Temp) = 0.97 - { [(2/5) * Info(S_{Hot})] + [(2/5) * Info(S_{Mild})] + [(1/5) * Info(S_{Cool})] }
- Gain (S, Temp) = 0.570

- **Values(Temp) = {Hot, Mild, Cool}**

- **S = { 2 , 3 }**
- **S_{Hot} = { 0 , 2 }**
- **S_{Mild} = { 1 , 1 }**
- **S_{Cool} = { 1 , 0 }**

- **Info(S) = [- (2/5) $\log_2$(2/5)] + [- (3/5) $\log_2$(3/5)] = 0.97**
- **Info(S_{Hot}) = 0.0**
- **Info(S_{Mild}) = 1.0**
- **Info(S_{Cool}) = 0.0**

Day	Temp	Humidity	Wind	Play Tennis
Dl	Hot	High	Weak	No
D2	Hot	High	Strong	No
D8	Mild	High	Weak	No
D9	Cool	Normal	Weak	Yes
Dl1	Mild	Normal	Strong	Yes

Day	Temp	Humidity	Wind	Play Tennis
Dl	Hot	High	Weak	No
D2	Hot	High	Strong	No
D8	Mild	High	Weak	No
D9	Cool	Normal	Weak	Yes
Dl1	Mild	Normal	Strong	Yes

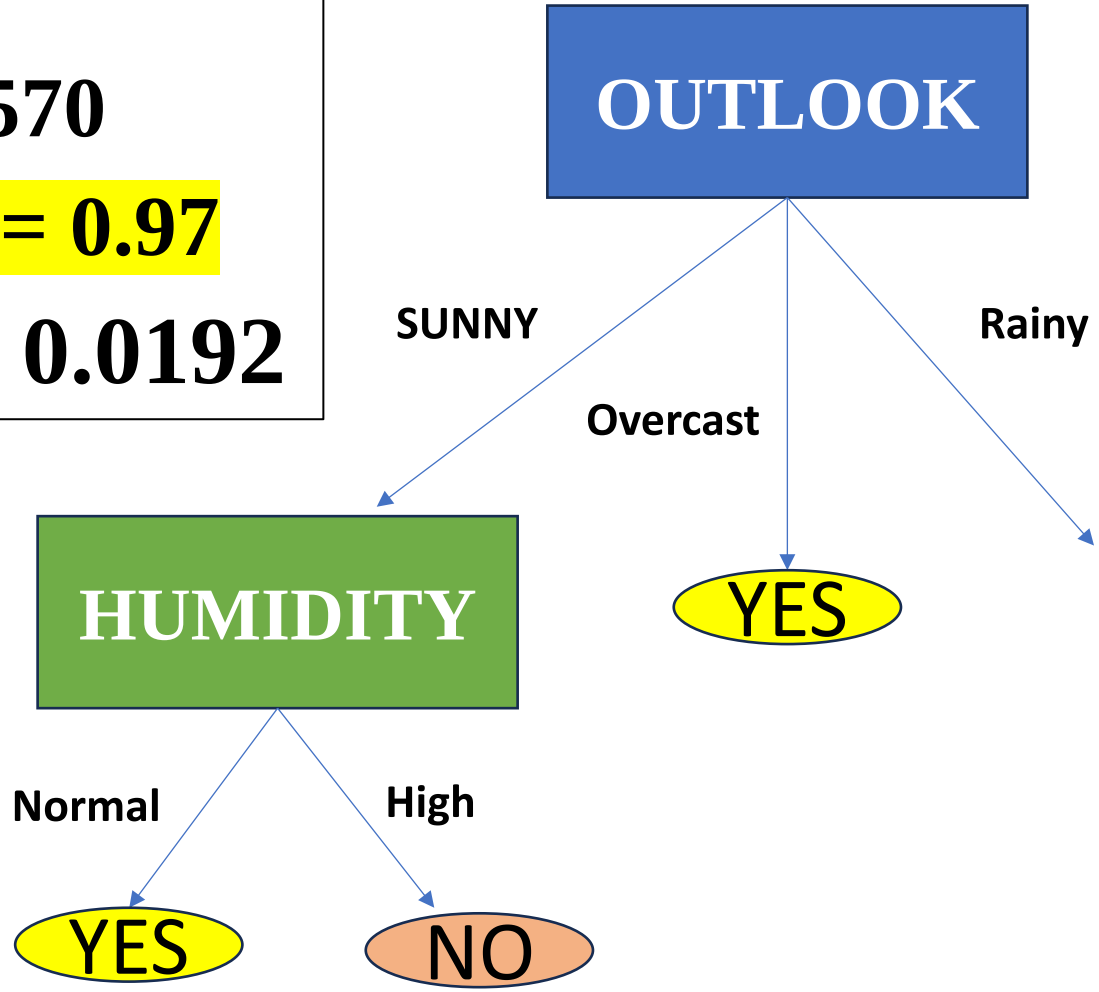
Calculate Gain for Humidity

- **Info(S) = 0.97**
- **Info(S_{High}) = 0.0**
- **Info(S_{Normal}) = 0.0**
- **Gain (S_{Sunny} , Humidity) = 0.97**

Calculate Gain for Wind

- **Info(S) = 0.97**
- **Info(S_{Weak}) = 0.9182**
- **Info(S_{Strong}) = 1.0**
- **Gain (S_{Sunny} , Wind) = 0.0192**

- Gain (S_{sunny} , Temp) = 0.570
- Gain (S_{Sunny} , Humidity) = 0.97
- Gain (S_{Sunny} , Wind) = 0.0192



Dataset for Rainy

Day	Temp	Humidity	Wind	Play Tennis
D4	Mild	High	Weak	Yes
D5	Cool	Normal	Weak	Yes
D6	Cool	Normal	Strong	No
D10	Mild	Normal	Weak	Yes
D14	Mild	High	Strong	No

Calculate Gain for Humidity

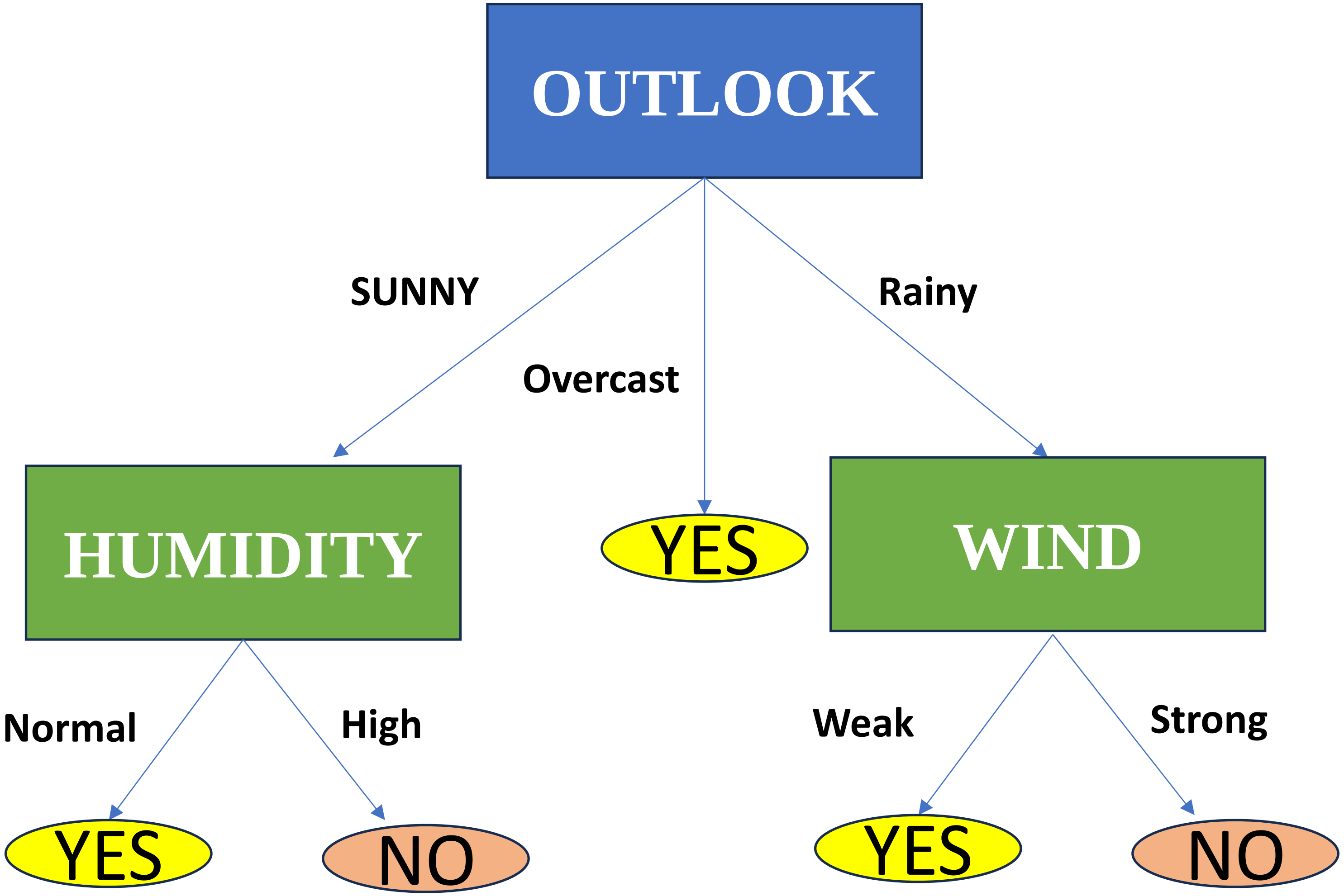
- **Gain (S_{Rain} , Humidity) = 0.0192**

Calculate Gain for Temp

- **Gain (S_{Rain} , Temp) = 0.0192**

Calculate Gain for Wind

- **Gain (S_{Rain} , Wind) = 0.97**


OUTLOOK
SUNNY
Overcast
Rainy
HUMIDITY
YES
WIND
Normal
High
Weak
Strong
YES
NO
YES
NO

A) Draw Decision Tree using ID3 algorithm for the following dataset. **(8)**

Hair	**Height**	**Weight**	**Class**
Blonde	Average	Light	No
Blonde	Tall	Average	Yes
Brown	Short	Average	Yes
Blonde	Short	Average	No
Red	Average	Heavy	No
Brown	Tall	Heavy	No
Brown	Average	Heavy	No
Blonde	Short	Light	Yes

b) Construct the decision tree using ID3 algorithm, for predicting the target attribute "Weather". 10

State	**Season**	**Barometer**	**Weather**
AK	Winter	Down	Snow
HI	Winter	Down	Sun
HI	Summer	Up	Sun
CA	Summer	Up	Rain
AK	Winter	Up	Snow
CA	Winter	Down	Sun
AK	Summer	Down	Sun
CA	Winter	Up	Rain
HI	Summer	Down	Sun

A) Draw Decision Tree for the following dataset. (Assume attribute, annual income into three categories: **<80, 80…99 and >=100**) **(10)**

TID	Home Owner	Marital Status	Annual Income	Class:Loan Defaulter
1.	Y	S	125	N
2.	N	M	100	N
3.	N	S	70	N
4.	Y	M	120	N
5.	N	D	95	Y
6.	N	M	60	N
7.	Y	D	220	N
8.	N	S	85	Y
9.	N	M	75	N
10.	N	S	90	Y

A) Draw Decision Tree for the following dataset. (10)

TID	Home Owner	Marital Status	Annual Income	Class: Loan Defaulter
1.	Y	S	high	N
2.	N	M	high	N
3.	N	S	low	N
4	Y	M	high	N
5.	N	D	Medium	Y
6.	N	M	low	N
7.	Y	D	high	N
8.	N	S	Medium	Y
9.	N	M	low	N
10.	N	S	Medium	Y